

Part I

SUMMARY of Related Rates

Summary

February 21, 2026

0.1 General steps for solving Related Rates problems.

- Introduce variables, identify the given rate and unknown rate.
- Draw a picture.
- Find equations.
- Differentiate with respect to t .
- Evaluate and solve.

Part II

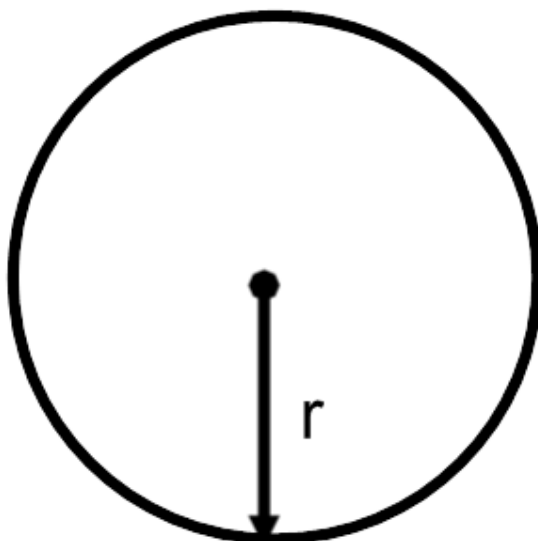
Recitation Questions

Problem 1

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Problem 1

Problem 1 The radius of a circle is increasing at a rate of 2 inches per minute.



- (a) At what rate is the circumference of the circle changing when the radius is 10 inches?

Explanation. We know: $\frac{dr}{dt} = 2$ inches per minute and we want to find $\frac{dc}{dt}$ when $r = 10$.

$$\begin{aligned}c &= 2\pi r \\ \frac{dc}{dt} &= 2\pi \frac{dr}{dt} \\ \left[\frac{dc}{dt} \right]_{r=10} &= 2\pi(2) \\ \left[\frac{dc}{dt} \right]_{r=10} &= 4\pi \text{ inches per minute}\end{aligned}$$

- (b) At what rate is the area of the circle changing when the radius is 12 inches?

Explanation. We know: $\frac{dr}{dt} = 2$ inches per minute and we want to find $\frac{dA}{dt}$ when $r = 12$.

Problem 1

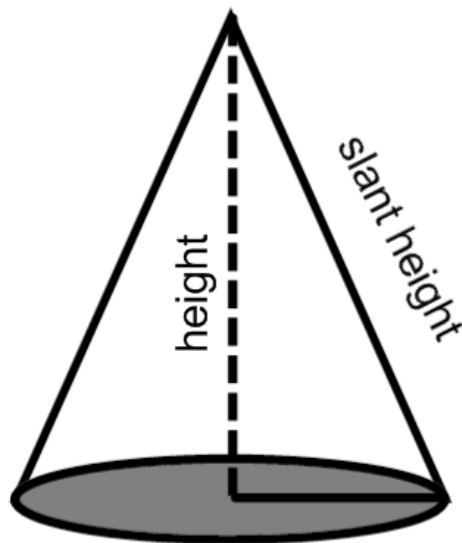
$$\begin{aligned}A &= \pi r^2 \\ \frac{dA}{dt} &= 2\pi r \frac{dr}{dt} \\ \left[\frac{dA}{dt} \right]_{r=12} &= 2\pi(12)(2) \\ \left[\frac{dA}{dt} \right]_{r=12} &= 48\pi \text{ inches squared per minute}\end{aligned}$$

Problem 2

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Problem 2

Problem 2 A right cone has a fixed slant height (see figure below) of 9 ft. The cone's height is shrinking at a rate of 0.5 ft/sec. At what rate is the **volume** of the cone changing when the height is 6 ft? Be sure to label the picture.



Explanation. We introduce the variables V , for the volume, r for the radius

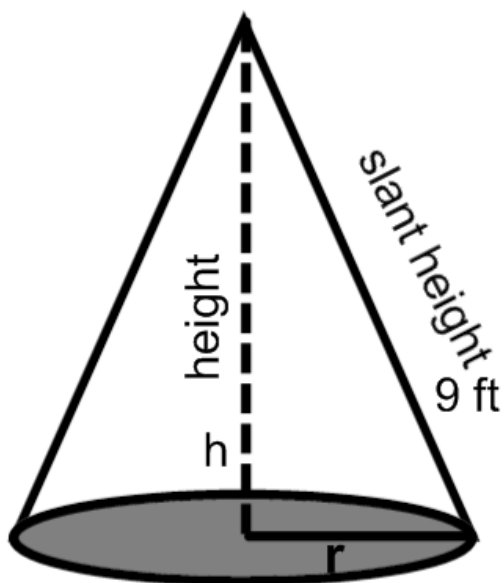
and h for the height of the cone.

The given rate is $\frac{dh}{dt} = -0.5$ ft/s;

the rate to be determined is $\left[\frac{dV}{dt} \right]_{h=6}$.

We label the picture.

Problem 2



The equation relating all relevant variables is

$$V = \frac{1}{3}\pi r^2 \cdot h$$

Before we differentiate, we will express r^2 in terms of h .

$$r^2 = 81 - h^2$$

and we substitute this into the equation above

$$V = \frac{1}{3}\pi(81 - h^2) \cdot h$$

Then we simplify it.

$$V = \frac{1}{3}\pi(81h - h^3)$$

Now, we differentiate with respect to t .

$$\frac{dV}{dt} = \frac{\pi}{3}(81 - 3h^2) \cdot \frac{dh}{dt}$$

Problem 2

Now, we evaluate.

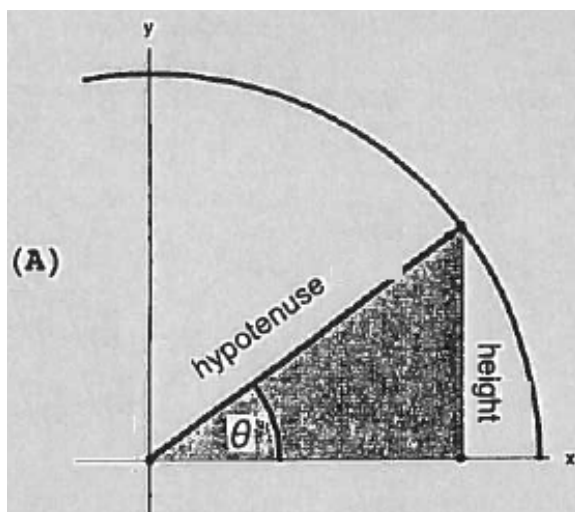
$$\begin{aligned}\left[\frac{dV}{dt}\right]_{h=6} &= \frac{\pi}{3}(81 - 3(6)^2)(-0.5) \\ \left[\frac{dV}{dt}\right]_{h=6} &= \frac{\pi}{3}(-27)(-0.5) \\ \left[\frac{dV}{dt}\right]_{h=6} &= 4.5\pi \text{ ft}^3/\text{s}\end{aligned}$$

Problem 3

February 21, 2026

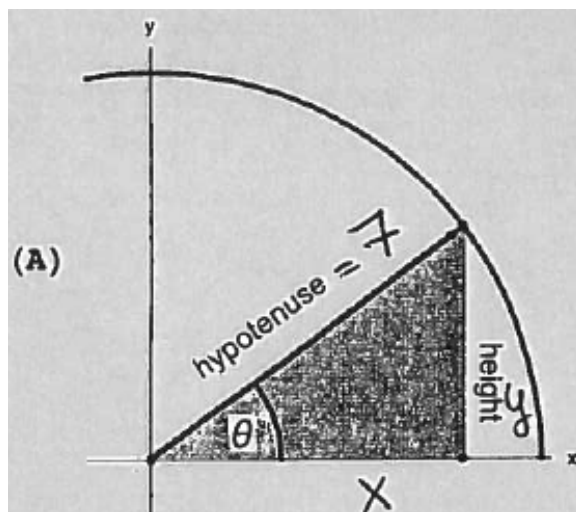
Problem 3

Problem 3 A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see Figure (A)). One of its sides lies on the x -axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x -axis. Assume that the angle θ changes at the rate $\frac{d\theta}{dt} = 0.2$ radians per second.



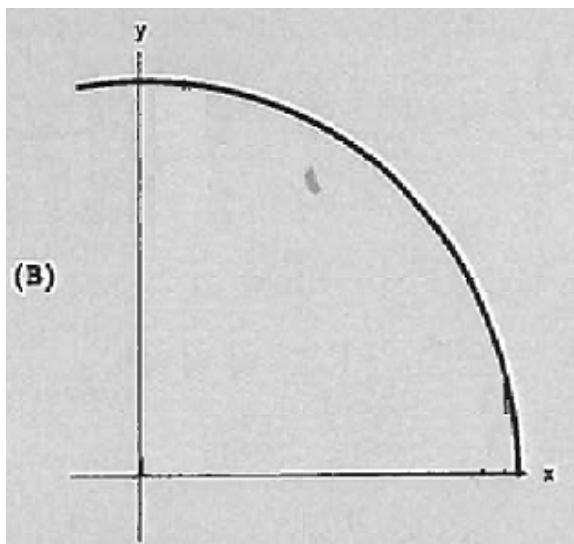
(a) Label Figure A.

Explanation.

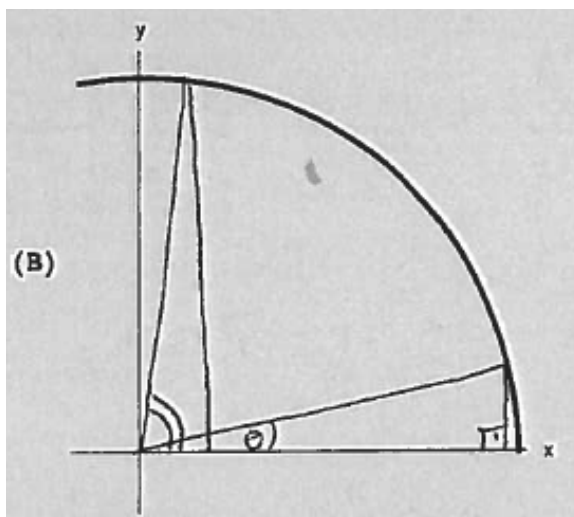


Problem 3

- (b) In Figure B, draw the triangle twice; once when θ is small and once more, when θ is close to $\frac{\pi}{2}$.



Explanation.



- (c) Find the rate of change of the height of the triangle when $\theta = \frac{\pi}{3}$

Explanation. $y = 7 \sin \theta$

$$\begin{aligned}\frac{dy}{dt} &= \frac{dy}{d\theta} \cdot \frac{d\theta}{dt} \\ \frac{dy}{dt} &= 7 \cos \theta \cdot \frac{d\theta}{dt} \\ \left[\frac{dy}{dt} \right]_{\theta=\frac{\pi}{3}} &= 7(1/2)(0.2) \\ &= 0.7 \text{cm/sec}\end{aligned}$$

- (d) Find the rate of change of the area of the triangle when $\theta = \frac{\pi}{3}$

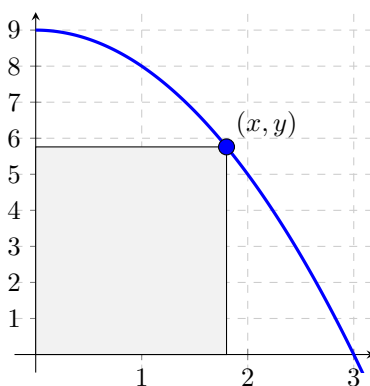
Explanation. First, we need to find x : $x = 7 \cos \theta$. From part c, we know $y = 7 \sin \theta$

$$\begin{aligned}A &= \frac{1}{2}xy \\ A &= \frac{49}{2} \cos(\theta) \sin(\theta) \\ \frac{dA}{dt} &= \frac{49}{2} (-\sin^2(\theta) + \cos^2(\theta)) \cdot \frac{d\theta}{dt} \\ \left[\frac{dA}{dt} \right]_{\theta=\frac{\pi}{3}} &= \frac{49}{2} \left(-\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right) \cdot 0.2 \\ &= \frac{49}{2} \cdot \frac{-1}{2} \cdot 0.2 \\ &= \frac{-4.9}{2} \text{cm}^2/\text{sec}\end{aligned}$$

Problem 4

February 21, 2026

Problem 4 A rectangle in the first quadrant is constructed by taking a point (x, y) on the graph of the function $f(x) = 9 - x^2$, drawing a line segment vertically downward to the x -axis and a line segment horizontally leftward to the y -axis, as in the picture below. Denote the **length of the base** (along the x -axis) of the rectangle by x , the **height of the rectangle** (along the y -axis) by y (in meters), and the **PERIMETER** of the rectangle by P . When $x = 2$ m (and only at that moment), the height y is shrinking at a rate of $\frac{1}{5}$ m/s. Find the value of $\left[\frac{dP}{dt} \right]_{x=2m}$ by performing the steps below.



- (a) **Find a formula** for the height, y , of the rectangle as a function of x . (This is denoted as $y(x)$.)

Explanation. The height of the rectangle is the y -coordinate of the point in the upper-right corner. Since this point lies on the graph of $f(x) = 9 - x^2$, we know $y = 9 - x^2$.

- (b) **Find the value** of $\left[\frac{dx}{dt} \right]_{x=2m}$.

Explanation. Since $y = 9 - x^2$, we know $\frac{dy}{dt} = -2x \frac{dx}{dt}$. Solving for $\frac{dx}{dt}$

gives:

$$\begin{aligned}\frac{dx}{dt} &= -\frac{\frac{dy}{dt}}{2x} \\ \left[\frac{dx}{dt}\right]_{x=2m} &= -\frac{\left[\frac{dy}{dt}\right]_{x=2m}}{2(2)} \\ &= -\frac{(-1/5)}{4} \\ &= \frac{1}{20}\end{aligned}$$

- (c) **Find a formula** for the perimeter, P , of the rectangle as a function of x . (This is denoted as $P(x)$.)

Explanation.

$$\begin{aligned}P(x) &= 2x + 2y \\ &= 2x + 2(9 - x^2) \\ &= -2x^2 + 2x + 18\end{aligned}$$

- (d) **Find the value** of $\left[\frac{dP}{dt}\right]_{x=2m}$.

Explanation.

$$\begin{aligned}P(x) &= -2x^2 + 2x + 18 \\ \frac{dP}{dt} &= (-4x + 2)\frac{dx}{dt} \\ eval \frac{dP}{dt} \bigg|_{x=2m} &= \left[(-4x + 2)\frac{dx}{dt}\right]_{x=2m} \\ &= (-4(2) + 2)\left(\frac{1}{20}\right) \\ &= -\frac{3}{10}\end{aligned}$$

- (e) **Write a sentence to explain** what the value found in (d) means about the rectangle. (Don't forget UNITS.)

Explanation. At the instant when $x = 2$ m, the perimeter of the rectangle is shrinking at the rate of $\frac{3}{10}$ m/s.